

Anchoring of stated preferences after policy implementation

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Abstract

This paper examines whether stated willingness to pay (WTP) changes when a policy-relevant value becomes an actual, implemented price. Using two survey waves conducted immediately before and after an increase in conservation entrance fees, we compare stated WTP across periods and tourist groups. WTP increases significantly following implementation among domestic tourists, shifting toward the new fee, while no significant change is observed among foreign tourists. Findings suggest that policy implementation can alter the reference environment in which WTP is elicited. Researchers should consider the timing of valuation relative to policy implementation when interpreting WTP estimates for welfare analysis and policymaking.

Keywords: Stated preferences, willingness to pay, anchoring, public policy, entrance fees, protected areas

JEL Classification: D91, Q51, Q57

1 Introduction

Individual preferences are fundamental to economic analysis; however, their elicitation through stated-preference (SP) methods can exhibit systematic response patterns that depart from the predictions of rational choice theory. One important deviation is anchoring, whereby valuations depend on numerical reference points rather than being formed independently (Tversky & Kahneman, 1974). Monetary values introduced during the elicitation process can anchor stated willingness to pay (WTP) and affect resulting welfare estimates (Herriges & Shogren, 1996). This raises an important question for non-market valuation: if reference points shape stated WTP, can policy implementation itself alter the welfare measure obtained from SP surveys? This study examines whether stated WTP for access to a public good changes when a proposed policy value becomes an actual, implemented price.

Evidence on the persistence of anchoring effects is mixed. Anchoring can persist despite experience, while preference learning may reduce it as respondents become more familiar with the valuation task and the good being valued (Bateman et al., 2008). More recent evidence shows that policy-relevant values can anchor stated preferences (Arceneaux & Nicholson, 2024), and that even minimally informative anchors can shift stated policy attitudes (Broscious et al., 2024). However, this evidence largely concerns anchors introduced within surveys through hypothetical or proposed policy scenarios. Much less is known about whether an actual policy implementation can change the reference environment in which valuations are formed. If implemented prices become reference points for WTP, pre- and post-policy valuations may differ and need not provide policy-invariant measures of welfare.

We address this question by exploiting a substantial change in conservation entrance fees in the Galápagos National Park that affected domestic and foreign tourists differently. In August 2024, the entrance fee increased from US\$6 to US\$30 for domestic tourists (a 400% increase) and from US\$100 to US\$200 for foreign tourists (a 100% increase). Using exactly the same face-to-face surveys conducted immediately before and after the policy implementation, we test whether stated WTP changed when the new fee moved from a proposed price to the actual price of access. Importantly, the new amount was disclosed explicitly to respondents before eliciting their WTP

in both waves. We find a substantial increase in WTP among domestic tourists following policy implementation, with the distribution shifting toward the new US\$30 fee, whereas WTP among foreign tourists does not change significantly.

While our findings are consistent with implemented policy prices becoming reference points for subsequent valuations, this does not necessarily imply that the resulting WTP estimates are invalid measures of welfare. Behavioral economics suggests that patterns traditionally characterized as SP anomalies may instead reflect predictable forms of decision-making that are context-dependent and also occur outside hypothetical valuation settings (Lades et al., 2025). Researchers should therefore consider the timing of valuation relative to policy implementation and the reference prices respondents face when interpreting WTP estimates for welfare analysis and policymaking.

2 Data and empirical strategy

Data were collected from two cross-sectional survey waves conducted in the departure area of Baltra Airport, the main gateway to the Galápagos Islands. The increase of the entrance fee was implemented in August 1, 2024. The first wave was conducted in July 2024, immediately before the entrance fee change took effect, while the second wave was conducted right after the implementation, in August 2024. Exactly the same questionnaire and sampling procedure were used in both waves, conducted within a short time interval to facilitate comparisons across periods while limiting changes in external conditions. Tourists were selected using systematic sampling and completed the questionnaire in Spanish or English. The final sample consists of 340 domestic and foreign tourists.¹

Stated willingness to pay (WTP) was elicited using an open-ended elicitation question. While this format has limitations, it avoids introducing starting values within the elicitation question itself, which is particularly important in this study because our objective is to examine whether policy implementation affects stated WTP. After providing informed consent, respondents (aged 18 or older) were first asked whether they were aware of recent changes in entrance fees, providing a measure of *ex-ante* policy awareness. They were then informed of the new fee levels, described

¹From an initial sample of 349 respondents, three zero WTP responses and six incomplete questionnaires were excluded from the econometric analysis.

as proposed in the pre-policy wave and as implemented in the post-policy wave, before WTP was elicited. Therefore, respondents in both waves received information about the new fee amounts, but only those surveyed after the policy change faced them as implemented prices. Controlling for *ex-ante* policy awareness helps distinguish the association with policy implementation from differences attributable to prior knowledge of the new fee levels.

To estimate the change in WTP following policy implementation and its heterogeneity across tourist groups, we model individual valuations using a generalized linear model (GLM). WTP_i denotes the stated willingness to pay of individual i , and X_i is a vector of contextual and individual characteristics. After excluding zero responses, WTP is strictly positive and right-skewed; therefore, we model it using a Gamma distribution with a log link function. The conditional mean is specified in Eq. (1):

$$E(WTP_i | X_i) = \exp(X_i\beta) \quad (1)$$

This specification ensures positive predictions, taking into account the heteroskedasticity commonly observed in WTP data. To capture the effect of the policy change and its heterogeneity across tourist groups, we estimate (Eq. 2):

$$\log(\mu_i) = \beta_0 + \beta_1 Post_i + \beta_2 Domestic_i + \beta_3 (Post_i \times Domestic_i) + X_i' \gamma \quad (2)$$

where $\mu_i = E(WTP_i | X_i)$. The indicator $Post_i$ equals one for observations collected after implementation of the fee increase and zero for those collected before, while $Domestic_i$ identifies domestic tourists, with foreign tourists as the baseline category. The interaction term captures whether the change in WTP between the pre- and post-policy periods differs between domestic and foreign visitors. The vector X_i includes age, income, education, gender, environmental attitudes, and *ex-ante* awareness of the policy change. These control variables account for observed heterogeneity across respondents, allowing us to estimate differences in conditional mean WTP before and after policy implementation while accounting for observed individual characteristics.

Under the log-link specification, coefficients represent effects on the log conditional mean of WTP. In particular, β_1 captures the post-policy change for foreign tourists, while β_3 captures

the additional post-policy change for domestic tourists relative to foreign tourists. We use post-estimation margins to facilitate interpretation, reporting predicted mean WTP and changes across groups in monetary terms. We also assess robustness to extreme values by re-estimating the models using a trimmed WTP variable that excludes the upper tail of the WTP distribution.

3 Results

Table A.1 reports sample characteristics before and after the policy change. The two survey waves are broadly comparable, with no statistically significant differences in most characteristics. Age and *ex-ante* policy awareness are the only variables that differ across periods at the 5% significance level. We control for these variables in alternative specifications.

Table 1 reports descriptive statistics and tests for differences in means and distributions. Among domestic tourists, mean WTP increases from USD 17.9 before the policy change to USD 31.9 after implementation (+78.6%), while the median rises from USD 10 to USD 20. A t-test indicates that the difference in means is significant at the 5% level. By contrast, mean WTP among foreign tourists decreases from USD 192.4 to USD 173.9, while the median remains unchanged at USD 200; the difference in means is not statistically significant. We additionally compare the distributions using the nonparametric Kolmogorov-Smirnov (KS) test. The test indicates a statistically significant difference between the pre- and post-policy distributions for domestic tourists at the 1% level, while no significant difference is found for foreign tourists. Figure 1 illustrates this pattern, among domestic tourists, the WTP distribution shifts to the right after implementation, around USD 30, the new entrance fee. In contrast, the distribution for foreign tourists remains broadly stable across periods.

Table 2 reports estimates from the Gamma GLM models. Across specifications, the interaction between the post-policy period and domestic tourists is positive and statistically significant, indicating that the post-policy change in WTP differs systematically between domestic and foreign visitors. This result is robust to the inclusion of covariates (specification 2), the addition of *ex-ante* policy awareness (specification 3), and the exclusion of potential extreme WTP values (specifications 4-6). The coefficient on the post-policy variable is negative and statistically insignificant,

indicating no statistical change in WTP for foreign tourists post-policy implementation. The positive interaction term, by contrast, indicates a significantly larger post-policy change for domestic tourists relative to foreign visitors. *Ex-ante* policy awareness is not statistically significant.

Table 1: Descriptive statistics of tourists' willingness to pay

	Pre-policy		Post-policy		Δ	$\Delta\%$	T-test Ho: Post - Pre = 0
	Mean (SD)	Median	Mean (SD)	Median			
Domestic	17.9 (20.9)	10	31.9 (53.1)	20	14.1	+78.6%	$t = 2.154^{**}$
Foreign	192.4 (103.9)	200	173.91 (85.3)	200	-18.5	-9.6%	$t = -1.124$

Notes: WTP is stated willingness to pay in USD. SD denotes standard deviation. The t-test assesses differences in mean WTP between periods. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

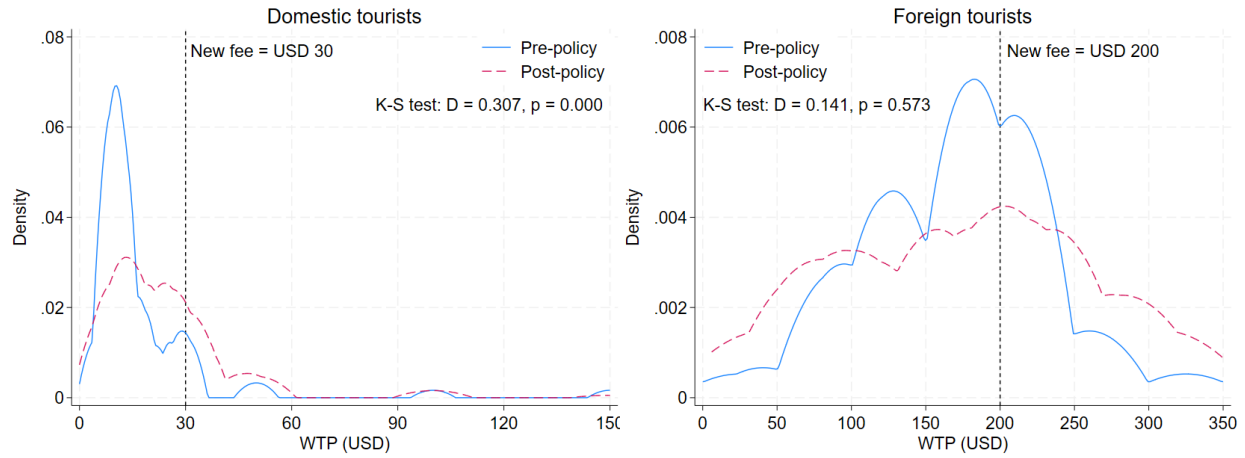


Figure 1: Distribution of stated WTP before and after policy implementation

Notes: The figure shows kernel density distributions of stated willingness to pay (WTP) among foreign (left) and domestic (right) tourists before and after policy implementation. The Kolmogorov-Smirnov (KS) test assesses whether the pre- and post-policy distributions differ. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To facilitate interpretation in monetary terms, Table 3 reports predicted WTP from specification (3), which results are strongly in line with those presented in Table 1. For foreign tourists, predicted WTP declines slightly, but the change is not statistically significant. For domestic tourists, predicted WTP increases from approximately USD 19.6 before the policy change to USD 33.6 after implementation, a significant increase of about USD 13.9. Even at the lower bound of the 95% confidence interval, the estimated increase is economically significant, it accounts for USD 4.8, equivalent to 24% of predicted pre-policy WTP. When potential extreme values are excluded, the predicted WTP among domestic tourists post-policy implementation remains significantly higher than before implementation, at approximately USD 26.7. Both the untrimmed and trimmed post-policy predictions are close to the new entrance fee of USD 30.

Overall, the results show a shift in stated WTP post-policy implementation among domestic tourists, but not among foreign visitors. While Broschius et al. (2024) show that even minimally informative anchors can influence stated preferences, and Arceneaux and Nicholson (2024) find anchoring effects from policy-relevant values associated with the status quo or policy proposals, our findings suggest that policy implementation itself can shape stated WTP. Importantly, this shift cannot be attributed simply to disclosure of the new fee, as respondents were informed of its value before WTP elicitation in both waves, nor is it explained by *ex-ante* awareness of the policy change. The observed pattern is therefore consistent with anchoring as described by Tversky and Kahneman (1974), but suggests that the new fee became a reference point for domestic tourists once implemented as the actual price of access. The asymmetric response between domestic and foreign tourists may reflect differences in baseline fees, the magnitude of the fee increase, or the fee relative to total travel expenditure. These mechanisms cannot be separately identified in our setting and deserve further research.

Table 2: Tourists' willingness-to-pay estimates - Gamma GLM with log link

	(1) Baseline	(2) Baseline +Cov.	(3) Baseline ++Cov.	(4) Trimmed	(5) Trimmed +Cov.	(6) Trimmed ++Cov
Post	-0.123 (0.091)	-0.086 (0.098)	-0.176 (0.141)	-0.123 (0.091)	-0.078 (0.093)	-0.148 (0.123)
Domestic	-2.384*** (0.156)	-2.194*** (0.151)	-2.198*** (0.150)	-2.384*** (0.156)	-2.148*** (0.134)	-2.154*** (0.132)
Post $\times$ Domestic	0.697*** (0.219)	0.622*** (0.190)	0.618*** (0.190)	0.421** (0.180)	0.381** (0.156)	0.380** (0.156)
Female		0.198* (0.102)	0.209** (0.101)		0.033 (0.081)	0.041 (0.077)
Higher education		0.264 (0.167)	0.245 (0.160)		0.226* (0.116)	0.217* (0.114)
Age		-0.058* (0.035)	-0.059* (0.035)		-0.044 (0.029)	-0.046 (0.029)
Income		0.081 (0.054)	0.086* (0.052)		0.115*** (0.042)	0.118*** (0.041)
Ex-ante awareness		-0.081 (0.109)	-0.216 (0.154)		-0.074 (0.084)	-0.177 (0.150)
Post $\times$ Ex-ante awareness			0.205 (0.201)			0.154 (0.172)
Constant	5.282*** (0.076)	5.050*** (0.187)	5.116*** (0.204)	5.282*** (0.076)	5.045*** (0.179)	5.098*** (0.203)
Observations	340	331	331	336	327	327
Log-likelihood	-1721.248	-1664.443	-1664.083	-1668.815	-1612.914	-1612.708
AIC	3450.496	3346.886	3348.167	3345.630	3243.828	3245.417

Notes: Robust standard errors in parentheses. Reference categories are pre-policy period, foreign tourist, male, and secondary education or lower. Models are estimated using a Gamma GLM with log link. Models (3) and (4) exclude extreme WTP values equal or above USD 200 for domestic tourists. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Predicted tourists’ willingness-to-Pay - USD per person

	WTP			WTP trimmed		
	Pre-policy	Post-policy	Difference	Pre-policy	Post-policy	Difference
Domestic	19.62 (2.42)	33.57 (3.95)	+13.95*** (4.66)	19.72 (2.35)	26.70 (2.19)	+6.99** (3.12)
Foreign	176.66 (17.82)	162.94 (11.74)	-13.71 (17.45)	169.94 (14.21)	157.39 (9.59)	-12.55 (15.92)

Notes: Predicted values from Gamma GLM with log link, using specification (3) for WTP and specification (6) for WTP trimmed; full estimates shown in Table 2. Robust standard errors in parentheses. Difference represents the absolute and relative change of the post-policy period relative to pre-policy. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4 Conclusion

This paper provides field evidence that stated willingness to pay (WTP) can change when a policy-relevant value becomes an actual, implemented price. While these findings may reflect anchoring or reference-dependent valuation, policy implementation may also change the credibility and informational content of the price. More broadly, policy implementation can modify the reference environment in which subsequent WTP is elicited. Such context dependence does not necessarily imply that pre- or post-implementation valuations are invalid, but it does affect their interpretation as welfare measures. Researchers and policymakers should therefore carefully consider the timing of elicitation and the reference prices respondents face when interpreting WTP estimates. Our analysis captures only short-run responses in one policy context; future research should examine whether policy-induced changes in stated valuations persist over time and generalize to other public goods and policy settings.

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A Appendix

Table A.1: Sample characteristics

Variable		Pre-policy	Post-policy	Ho: No difference p-value
Tourist nationality	Foreign	39%	43%	0.448
	Domestic	61%	57%	
Gender	Male	41%	47%	0.237
	Female	59%	53%	
Age 18 - 83 years	Mean	42.7	38.8	0.018
	(std. dev.)	(15.1)	(14.4)	
Education	Secondary or lower	23.3%	16.3%	0.111
	Higher education	76.7%	83.7%	
Monthly income USD 0.3k - 150k	Mean	12.5k	8.3k	0.182
	(std. dev.)	(33.0k)	(23.2k)	
Ex-ante policy awareness	Yes	48%	62%	0.018
	No	52%	38%	
Observations	N = 349	122	227	—

Notes: The null hypothesis (Ho) that there are not statistical differences across all group means is tested using analysis of variance (ANOVA).